

Final Assessment-Alternative Assessment

Semester	202520	Division	CIS
Assessment title in Syllabus	Final_Project_Portfolio	Program	Bachelor of Information Technology/ Bachelor of Information Systems

Course Code	CDS 2413		
Course Title	INTRODUCTION TO DATA SCIENCE		
Course Instructor		CRN	
Assessment	Project Deliverable 1 – Data Analysis Report	Submission Date	Week 6

Student Declaration:

Academic Integrity Statement

In accordance with the HCT Academic Integrity Policy

- Students are required to refrain from all forms of academic integrity breaches as defined and explained by HCT.
- A student found guilty of having committed acts of academic integrity breach(es) will be subject to the relevant sanctions as outlined by HCT.

إفادة النزاهة الأكاديمية

وفقاً لسياسة كليات التقنية العليا للنزاهة الأكاديمية

- على الطلبة الالتزام بلوائح وقواعد النزاهة الأكاديمية، كما هو مبين وموضح في السياسات والإجراءات الخاصة بكليات التقنية العليا.
- في حالة ارتكاب الطالب أي شكل من أشكال الإخلال بالنزاهة الأكاديمية، سيتعرض الى العقوبات الموضحة في السياسات ذات الصلة.

This assignment is entirely my/our own work except where I have duly acknowledged other sources in the text and listed those sources at the end of the assignment. I have not previously submitted this work to the HCT, or any other entity. I understand that I may be orally examined on my submission.

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Student Name(s):	Shama alharthi	Riham Alhosani	Khawla Alshehhi	
Student Signature				

Chatbot (e.g. ChatGPT) statement

This submission allows you to use AI generated tools, but the prompt used must be stated and the response must be in red font. Include References to ChatGPT.

Instructions

Remove instructions and hints from the template before submitting the final report to BBLearn.

CLO1: Introduction and Data Wrangling (5 Points)

Purpose of Data Analysis (1 point)

The reason for this analysis is to look at how global safety has changed from 1970 to 2025. By looking at the homicide rates, we can see if the world is getting safer over time. I also want to compare specific countries, like the USA and Japan, to the global average to see which places have the lowest crime rates and how they have improved over the years.

Type of Programming (1 point)

We will be using Python for this project because it is the best language for data science. It has special tools (libraries) like Pandas for organizing the data and Matplotlib for making the graphs. Using Python makes it much easier to handle large amounts of data and perform math tests quickly.

Type and Purpose of ML algorithm (1 point)

we plan to use Supervised Learning, specifically Linear Regression. The purpose of this is to predict future homicide rates. Since we have the years and the past rates, the algorithm can find a trend and help us guess what the global average might look like in the next few years.

Identify the Variables (1 point)

- Dependent Variable: Global_Avg. This is what we are measuring. It is called "dependent" because we want to see how it changes based on the year.
- Independent Variable: Year. This is the variable that we use to explain the change. Time moves forward regardless of the crime rate, so it is the independent factor.

Sampling (1 point)

Yes, we will use sampling to test the data.

- Type of Sampling: Random and Systematic sampling.
- Justification: we will use Random Sampling to pick 150 random data points to see if a small group represents the whole dataset correctly. we will also use Systematic Sampling by picking every 5th year from the list to see the trend at regular intervals without looking at every single year.

CLO2: Descriptive, Exploratory and Statistical Analysis (10 Points)

Purpose of Descriptive Analysis (1 point)

The Exam_Score variable in the dataset is summarized and comprehended using descriptive analysis. It assists in determining the distribution pattern, minimum and maximum values, quartiles, average score, and score spread. This is helpful because, prior to conducting more in-depth exploratory research and hypothesis testing, descriptive statistics provide a clear picture of student performance. Descriptive statistics are useful for summarizing data, comparing values, and identifying outliers, as was explained in Week 3.

Script for Descriptive Analysis (1 point)

```
import pandas as pd
df = pd.read_csv("StudentPerformanceFactors.csv")
def descriptive_stats(data, field):
    s = data[field].dropna()
    result = {
        "count": s.count(),
        "mean": s.mean(),
        "median": s.median(),
        "mode": s.mode().iloc[0],
        "std": s.std(),
        "variance": s.var(),
        "min": s.min(),
        "q1": s.quantile(0.25),
        "q3": s.quantile(0.75),
        "max": s.max(),
        "iqr": s.quantile(0.75) - s.quantile(0.25),
        "skewness": (s.mean() - s.median()) / s.std()
    }
    return result

print(descriptive_stats(df, "Exam_Score"))
```

Random Sampling(1 point)

A random sample of size 150 was selected from the dataset, and descriptive statistics were calculated for Exam_Score.

```
random_sample = df.sample(n=150, random_state=42)
print(descriptive_stats(random_sample, "Exam_Score"))
```

Random sample descriptive statistics for Exam_Score:

Count = 150
Mean = 67.41
Median = 67.00
Mode = 67
Standard Deviation = 4.12

Variance = 16.99

Minimum = 55

Q1 = 65

Q3 = 70

Maximum = 89

IQR = 5

Pearson's Coefficient of Skewness = 0.10

Interpretation : With a mean score of 67.41, the random sample is extremely near to the population mean. The distribution is roughly symmetric with a small positive skew because the skewness value is near zero.

Systematic Sampling (1 point)

Every 44th record in the dataset was chosen for systematic sampling, yielding a sample of roughly 151 observations.

```
systematic_sample = df.iloc[::44]
```

```
print(descriptive_stats(systematic_sample, "Exam_Score"))
```

Count = 151

Mean = 67.98

Median = 67.00

Mode = 66

Standard Deviation = 4.70

Variance = 22.09

Minimum = 61

Q1 = 65

Q3 = 70

Maximum = 94

IQR = 5

Pearson's Coefficient of Skewness = 0.21

Interpretation : Although it has a somewhat larger dispersion than the random sample, the systematic sample likewise has a mean that is near to the population mean. A slight right skew is indicated by the skewness, which is still tiny and positive.

Descriptive Statistical Report (1 point)

A detailed descriptive statistics report was prepared for the dependent variable Exam_Score.

Population descriptive statistics for Exam_Score:

Count = 6607

Mean = 67.24

Median = 67.00

Mode = 68

Standard Deviation = 3.89

Variance = 15.14

Minimum = 55

Q1 = 65

Q3 = 69

Maximum = 101

IQR = 4

Pearson's Coefficient of Skewness = 0.06

Report :

A typical examination result of 67.24 represents a generally decent level of student performance. The distribution appears to be almost symmetric, as indicated by the median of 67, which is extremely near to the mean. With a standard deviation of 3.89, the majority of student scores are close to the mean. The middle 50% of scores fall within a small range between 65 and 69, according to the interquartile range of 4. There are some extreme results, represented by the minimum score of 55 and the maximum score of 101. The even distribution is approximately symmetric with a very little positive skew because the Pearson's coefficient of skewness is 0.06. These metrics provide a clear summary of central tendency, dispersion, and position based on Week 3.

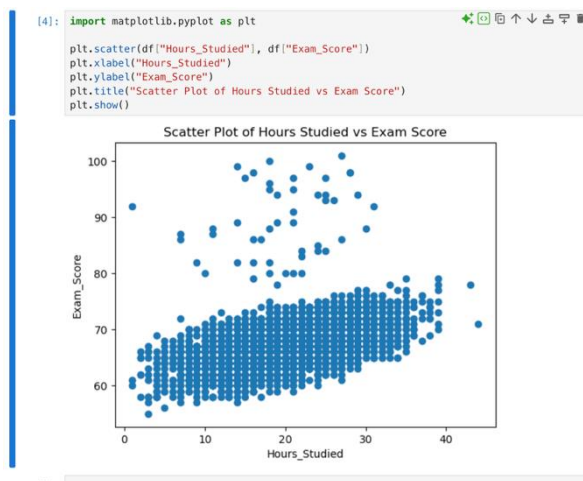
Visualisation through Various Charts (2 points)

```
import matplotlib.pyplot as plt

plt.scatter(df["Hours_Studied"], df["Exam_Score"])
plt.xlabel("Hours_Studied")
plt.ylabel("Exam_Score")
plt.title("Scatter Plot of Hours Studied vs Exam Score")
plt.show()
```

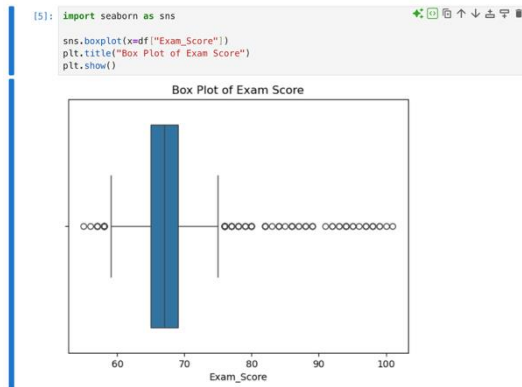
Improved Interpretation:

Exam_Score and Hours_Studied have a positive correlation, according to the scatter plot. Exam scores typically rise in tandem with an increase in study time. The association is good but not ideal because the points are evenly distributed.



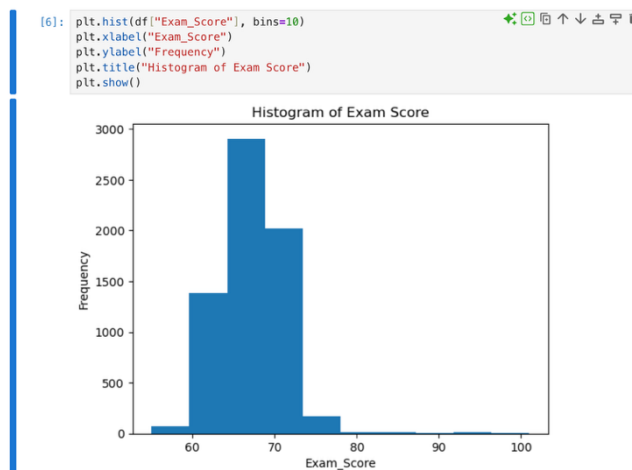
a. import seaborn as sns

```
sns.boxplot(x=df["Exam_Score"])
plt.title("Box Plot of Exam Score")
plt.show()
```



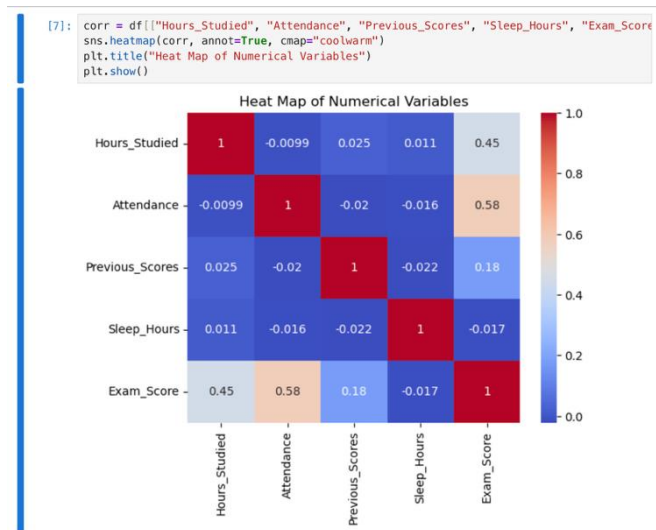
The majority of exam results appear in the center, as the box plot represents. The main 50% of scores fall around 65 and 69, with the average being approximately 67. Outliers may be indicated by a few high numbers that occur outside of the main range.

- b. `plt.hist(df["Exam_Score"], bins=10)`
`plt.xlabel("Exam_Score")`
`plt.ylabel("Frequency")`
`plt.title("Histogram of Exam Score")`
`plt.show()`



The majority of student scores are concentrated in the mid-to-high 60s, according to the histogram. Because there are some higher values on the right side, the distribution is somewhat right-skewed but otherwise almost normal.

- c. `corr = df[["Hours_Studied", "Attendance", "Previous_Scores", "Sleep_Hours", "Exam_Score"]].corr()`
`sns.heatmap(corr, annot=True, cmap="coolwarm")`
`plt.title("Heat Map of Numerical Variables")`
`plt.show()`



Correlations between numerical variables can be seen with the aid of the heat map. Exam_Score and Hours_Studied, Attendance, and Previous_Scores are positively correlated. Heat maps are helpful for seeing trends and connections between variables .

Hypothesis Test (1 point)

Based on Week 5 guidelines for numerical variables, Pearson and Spearman correlations were employed to examine the link between the independent variable Hours_Studied and the dependent variable Exam_Score.

- H0: There is no significant correlation between Hours_Studied and Exam_Score.
- H1: There is a significant correlation between Hours_Studied and Exam_Score.

```
[8]: from scipy.stats import pearsonr, spearmanr

pearson_r, pearson_p = pearsonr(df["Hours_Studied"], df["Exam_Score"])
spearman_r, spearman_p = spearmanr(df["Hours_Studied"], df["Exam_Score"])

print("Pearson:", pearson_r, pearson_p)
print("Spearman:", spearman_r, spearman_p)
```

Pearson: 0.4454549540752819 1.28635e-319
Spearman: 0.48095596242825545 0.0

- Pearson correlation = 0.445
- Pearson p-value < 0.001

- Spearman correlation = 0.481
- Spearman p-value < 0.001

Hours_Studied and Exam_Score have a somewhat positive connection in both tests, and both p-values are less than 0.05. As a result, we reject the null hypothesis and come to the conclusion that the number of study hours and exam score are statistically correlated.

One Sample T-test (2 points)

Using the Week 5 technique, a one-sample t-test was employed to determine if the random sample is representative of the population with respect to the dependent variable Exam_Score.

Hypotheses

- H0: The sample mean of Exam_Score is equal to the population mean.
- H1: The sample mean of Exam_Score is different from the population mean.

Population mean of Exam_Score:

- 67.24

Random sample mean of Exam_Score:

- 67.41

```
[9]: from scipy.stats import ttest_1samp, shapiro

population_mean = df["Exam_Score"].mean()
sample_scores = random_sample["Exam_Score"]

shapiro_stat, shapiro_p = shapiro(sample_scores)
t_stat, t_p = ttest_1samp(sample_scores, population_mean)

print("Shapiro-Wilk p-value:", shapiro_p)
print("One-sample t-test p-value:", t_p)

Shapiro-Wilk p-value: 1.6460794178919468e-05
One-sample t-test p-value: 0.6121703345045606
```

- Shapiro-Wilk p-value = 0.000016
- One-sample t-test statistic = 0.508
- One-sample t-test p-value = 0.612

The p-value of the t-test (0.612) is higher than 0.05 and we do not reject the null hypothesis. In general there's no significant difference between the sample average and the overall population average .

End of Documentation.